

Structural isomerism

Learning outcomes

At the end of this section you should be able to recognise isomers, including branched, straight-chain, position and functional group isomers.

Gasoline (petrol) is a mixture of hydrocarbons. However, the mixture has to be carefully blended to make sure that it contains the correct type of hydrocarbons. Longer, straight-chain hydrocarbons can cause an engine to undergo knocking, also known as auto-ignition. Knocking occurs when the mixture of gasoline and air is compressed and ignites prematurely. This can cause a loss of power and can also cause damage to the engine. The measure of a fuel's ability to resist knocking is known as its octane rating. Shorter, more branched hydrocarbons have higher octane ratings and longer, more straight-chain hydrocarbons have lower octane ratings. Next time you are at the gas (or petrol) station, have a look at the pumps and you will see the octane ratings for each type of fuel (**Figure 1**). A higher number means more resistance to auto-ignition and knocking, but it is usually more expensive.



Figure 1. Gasoline (petrol) has different octane ratings depending on its ability to resist pre-ignition.

Credit: marekuliasz, Getty Images

 More information for figure 1

The image shows a gasoline pump panel with three fuel options: Regular Unleaded, Mid Grade Unleaded, and Premium Unleaded. Each section displays the octane rating and price per gallon. Regular Unleaded has an octane rating of 85 and costs \$2.959 per gallon. Mid Grade Unleaded shows an octane rating of 87 with a price of \$3.059 per gallon. Premium Unleaded features an octane rating of 91, priced at \$3.159 per gallon. Each section includes a small rectangular display indicating that all prices include taxes, and there is a yellow button labeled 'PRESS' beneath each octane rating.

[Generated by AI]

The full structural formulas of two compounds, butane and 2-methylpropane, are shown in **Figure 2**. Study the two compounds for a few minutes and then think about the similarities and differences between the two compounds. Check your answer when you are ready.

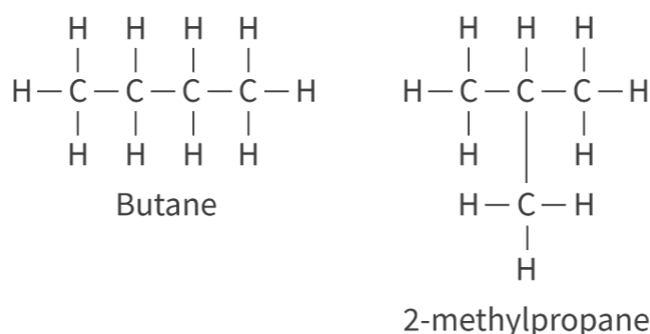


Figure 2. The full structural formulas of butane and 2-methylpropane.

 More information for figure 2

The image displays the full structural formulas of two compounds: butane and 2-methylpropane. On the left, butane is depicted as a straight chain of four carbon atoms (C) linked together, each bonded with hydrogen atoms (H), forming the structure $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3$. The connecting lines represent chemical bonds, and each carbon atom is fully saturated with hydrogen, showing a single bond with four hydrogens at the ends.

On the right, 2-methylpropane features a branched structure rather than a straight chain. The central carbon atom is connected to three other carbon atoms, forming a cross-like structure with hydrogen atoms filling the remaining bonds, specifically $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_3$. This arrangement makes it a branched alkane, distinguishing it from the linear structure of butane. Both structures have the same molecular formula, C_4H_{10} , but differ in their connectivity, illustrating isomerism.

[Generated by AI]

A count of the atoms in each compound reveals that they both have the same number of each type of atom, four carbon atoms and ten hydrogen atoms, but that the atoms are arranged differently. 2-methylpropane has a branch coming off the second carbon atom and is known as a branched-chain alkane, whereas propane is a straight-chain alkane. Compounds such as butane and 2-methylpropane are known as structural isomers, more specifically chain isomers. These compounds have the same molecular formulas but their parent chains have different lengths, with one or more of the isomers having branches or side-chains.

Next, we will consider two more organic compounds, but-1-ene and but-2-ene (**Figure 3**). What are the similarities and differences between the two compounds?

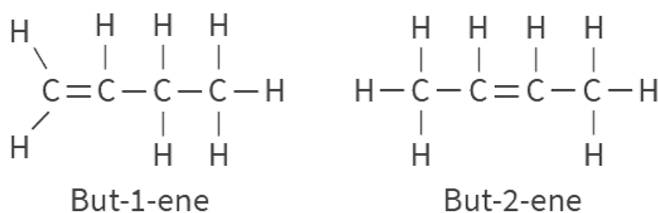


Figure 3. But-1-ene and but-2-ene.

 More information for figure 3

The image depicts the structural formulas of two organic compounds, but-1-ene and but-2-ene. On the left, the formula for but-1-ene is shown: a four-carbon chain with a double bond between the first and second carbons. Each carbon is bonded to the appropriate number of hydrogen atoms, with the carbon numbers 1 and 2 being bonded by a double bond. On the right, but-2-ene is displayed similarly, but the double bond is between carbon numbers 2 and 3. Both structures illustrate the position isomerism of these molecules, with identical numbers and types of atoms arranged differently.

[Generated by AI]

Both compounds have the same number and type of each atom, but the C=C bond is in a different position. But-1-ene has the C=C bond between carbon numbers 1 and 2 and but-2-ene between carbon numbers 2 and 3. These two compounds are also structural isomers, known as position isomers. Position isomers have the same molecular formula but the functional group is located in a different position.

Finally, we look at the last type of structural isomerism. **Figure 4** has the structures of two different compounds, propanal and propanone. Propanal is an aldehyde and propanone is a ketone. What is the same and what is different about the compounds? Check your answer below when you are ready.

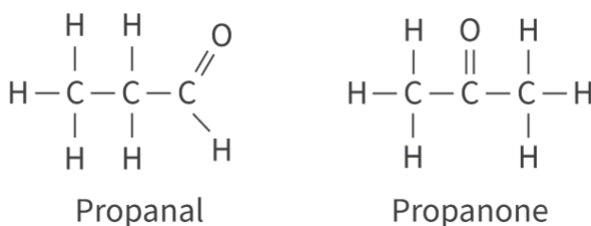


Figure 4. The full structural formulas of propanal and propanone.

 More information for figure 4

The image contains two structural formulas side by side. On the left is propanal, an aldehyde. It has a chain of three carbon atoms. The first carbon atom is bonded to three hydrogen atoms. The second carbon atom is bonded to two hydrogen atoms. The third carbon atom is double-bonded to an oxygen atom (carbonyl group) and also bonded to another hydrogen atom. Below this structure, the word 'Propanal' is written.

On the right is propanone, also known as acetone, which is a ketone. It also has a chain of three carbon atoms. The first carbon atom is bonded to three hydrogen atoms, similar to propanal. The second carbon atom is double-bonded to an oxygen atom (carbonyl group) and bonded to two other carbon atoms. The third carbon atom is bonded to three hydrogen atoms. Below this structure, the word 'Propanone' is written.

Both structures have the same number of carbon, hydrogen, and oxygen atoms, but they differ in the placement of the carbonyl group, which distinguishes the aldehyde from the ketone.

[Generated by AI]

Both compounds have the same molecular formula but the functional group is different. These are known as functional group isomers, which have the same molecular formula but a different functional group.

The different types of isomerism are summarised in **Figure 5**. Higher level (HL) students will study additional types of isomerism in section S3.2.7a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-cistrans-isomerism-hl-id-45666/>) and S3.2.7b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-optical-isomerism-hl-id-45667/>).

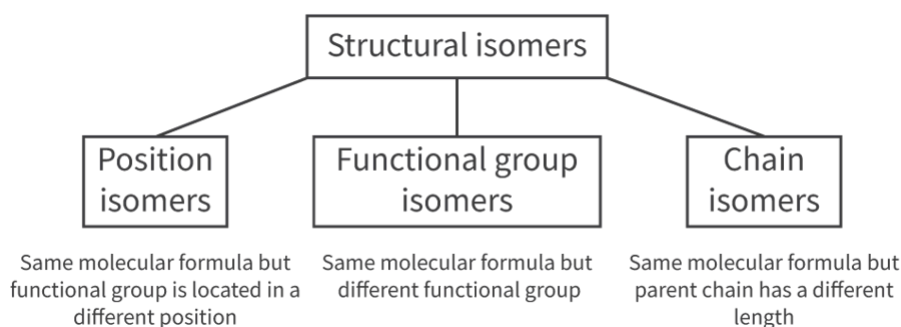


Figure 5. The different types of structural isomerism.

 More information for figure 5

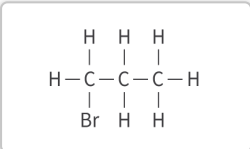
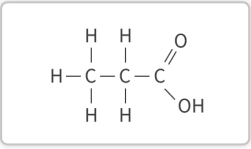
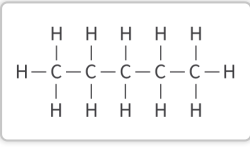
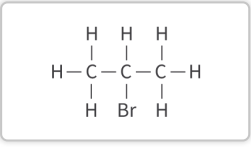
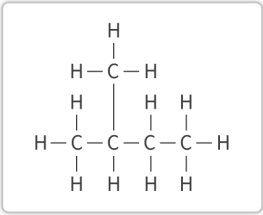
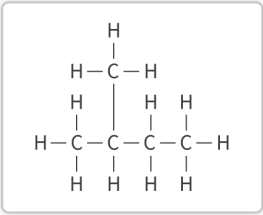
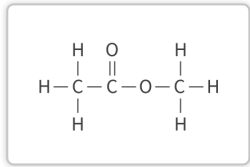
The diagram illustrates the different types of structural isomers by compartmentalizing them into three categories. At the top, a box labeled "Structural isomers" branches into three subcategories: Position isomers, Functional group isomers, and Chain isomers.

- Position isomers: These have the same molecular formula but differ in that the functional group is located in a different position.
- Functional group isomers: These share the same molecular formula but have a different functional group.
- Chain isomers: These also share the same molecular formula, but the parent chain has a different length.

Each type is demonstrated with a connecting line to highlight their relationship to the overarching category of structural isomers.

[Generated by AI]

Now try the exercise in **Interactive 1** by dragging the correct pairs of isomers into the correct boxes to test your understanding of structural isomers.

Chain isomers	Position isomers	Functional group isomers
		
		
		
☒ Check		

Interactive 1. Pairing Structural Isomers.

Primary, secondary and tertiary organic compounds

Organic compounds such as alcohols, amines and halogenoalkanes can be classified as primary, secondary or tertiary. This classification depends on the number of carbon atoms that are bonded to the carbon atom that is directly bonded to the functional group (or in the amines, the nitrogen atom of the amino group).

Figure 6 shows the three structural isomers of $C_4H_{10}O$. From left to right, we have butan-1-ol, butan-2-ol and 2-methylpropan-2-ol.

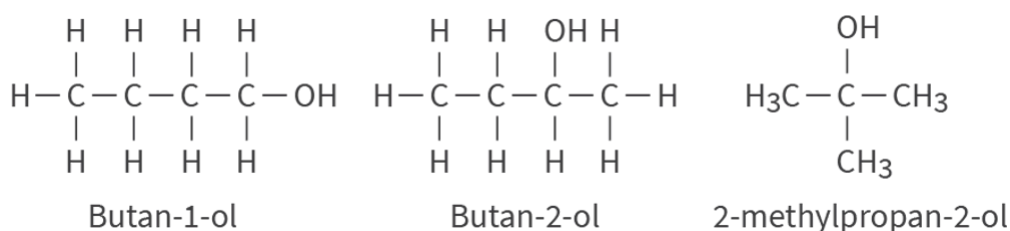


Figure 6. The full structural formulas of the alcohols, butan-1-ol, butan-2-ol and 2-methylpropan-2-ol.

 More information for figure 6

The image displays the structural formulas of three alcohols, each with the molecular formula $C_4H_{10}O$. From left to right:

1. Butan-1-ol:

2. Linear chain structure with four carbon atoms.
3. Hydroxyl group (OH) at the end of the chain.
4. Structure: $\text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{OH}$ with each carbon bonded to enough hydrogen atoms to complete four bonds per carbon.

5. Butan-2-ol:

6. Linear chain structure with four carbon atoms.
7. Hydroxyl group (OH) attached to the second carbon.
8. Structure: $\text{H} - \text{C} - \text{C} - \text{OH} - \text{C} - \text{C}$ with each carbon atom bonded to hydrogen atoms to form a complete structure.

9. 2-methylpropan-2-ol:

10. Branched structure.
11. Central carbon bonded to three other carbon groups and one hydroxyl group (OH).
12. Structure: $(\text{CH}_3)_3\text{C} - \text{OH}$ where the central carbon is connected to three CH_3 groups and an OH group.

These structural differences highlight how the position and branching affect the classification and properties of the alcohols.

[Generated by AI]

In butan-1-ol, the hydroxyl group is bonded to a carbon atom that is bonded to one other carbon. Therefore, butan-1-ol is a primary alcohol (it contains a primary carbon atom). In butan-2-ol the hydroxyl group is bonded to a carbon atom that is bonded to one hydrogen atom and two carbon atoms. For this reason, butan-2-ol is a secondary alcohol (it contains a secondary carbon atom). Finally, we have 2-methylpropan-2-ol in which the carbon atom bonded to the hydroxyl group is bonded to three other carbon atoms and no hydrogen atoms. 2-methylpropan-2-ol is classified as a tertiary alcohol (it contains a tertiary carbon atom).

Figure 7 summarises the structures of primary, secondary and tertiary alcohols. You may know that R' and R'' represent the alkyl groups that are bonded to the carbon atom, which can be the same or different.

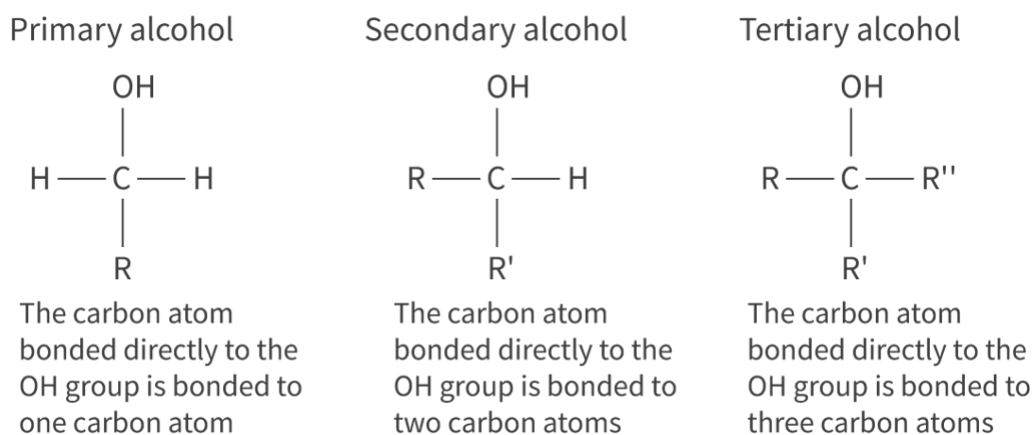


Figure 7. Alcohols can be classified as primary, secondary or tertiary.

 More information for figure 7

The image is a diagram showing the molecular structures of primary, secondary, and tertiary alcohols. Each structure is accompanied by a description.

1. **Primary Alcohol:** The structure has a central carbon (C) atom bonded to an OH group, a hydrogen (H) atom, and a generic alkyl group labeled as R. This type is classified based on the carbon atom bonded directly to the OH group being bonded to only one carbon atom.

2. **Secondary Alcohol:** In this structure, the central carbon is bonded to an OH group, an H atom, and two alkyl groups represented as R and R'. Here, the carbon atom bonded directly to the OH group is bonded to two other carbon atoms.
3. **Tertiary Alcohol:** This structure features a central carbon bonded to the OH group, and three alkyl groups labeled as R, R', and R". The carbon atom bonded directly to the OH group is connected to three carbon atoms.

The diagram illustrates how the number and nature of alkyl groups bonded to the central carbon determine the classification of the alcohol as primary, secondary, or tertiary.

[Generated by AI]

The halogenoalkanes can also be classified as primary, secondary or tertiary. **Figure 8** shows three examples of halogenoalkanes, one primary, one secondary and one tertiary. Can you deduce which one is which?

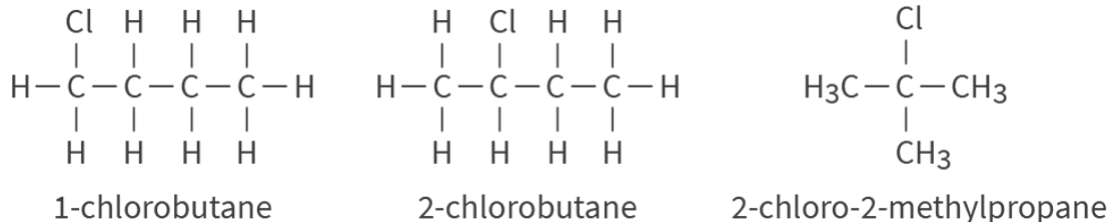


Figure 8. Can you deduce which of these molecules is a primary, secondary or tertiary halogenoalkane?

 More information for figure 8

The image displays three molecular structures of halogenoalkanes, each labeled with its chemical name. The first structure is 1-chlorobutane, which features a linear chain of four carbon atoms with a chlorine atom bonded to the first carbon. The second structure is 2-chlorobutane, showing a linear chain of four carbon atoms with a chlorine atom attached to the second carbon. The third structure is 2-chloro-2-methylpropane, with a central carbon atom bonded to three methyl groups and a chlorine atom, creating a branched configuration. Each molecule's hydrogen atoms are also displayed to complete the valency.

[Generated by AI]

1-chlorobutane is a primary halogenoalkane because the carbon atom bonded to the halogen is bonded to one other carbon and two hydrogens. 2-chlorobutane is a secondary halogenoalkane because the carbon atom bonded to the halogen is bonded to two other carbon atoms and one hydrogen. And 2-chloro-2-methylpropane is a tertiary halogenoalkane because the carbon atom bonded to the halogen is bonded to three carbon atoms and no hydrogen atoms.

Finally, we will look at how to classify the amines as either primary amines, secondary amines or tertiary amines. Amines are derivatives of ammonia in which the hydrogen atoms have been replaced with alkyl groups (**Figure 9**). If only one hydrogen atom has been replaced with an alkyl group, it is a primary amine. If two hydrogen atoms have been replaced, it is a secondary amine and if all three hydrogen atoms have been replaced with alkyl groups, it is a tertiary amine.

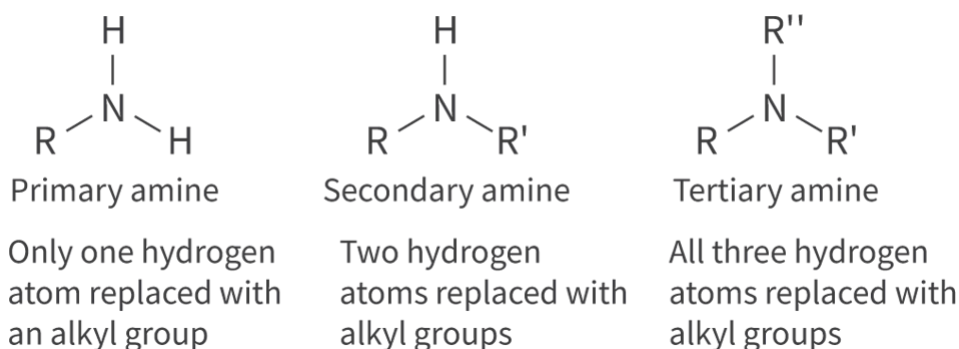


Figure 9. Amines are classified as primary, secondary tertiary depending on the number of alkyl groups bonded to the nitrogen atom of the amino group.

 More information for figure 9

The image is a diagram comparing primary, secondary, and tertiary amines through chemical structures.

1. **Primary amine:** The structure shows a nitrogen (N) atom bonded to one hydrogen (H) atom, another hydrogen, and an alkyl group (R). The accompanying text states: "Only one hydrogen atom replaced with an alkyl group."

2. **Secondary amine:** This structure has a nitrogen atom bonded to two different alkyl groups (R and R') and one hydrogen atom. The text indicates: "Two hydrogen atoms replaced with alkyl groups."
3. **Tertiary amine:** Here, the nitrogen is bonded to three different alkyl groups (R, R', and R"). The text says: "All three hydrogen atoms replaced with alkyl groups."

The diagram visually differentiates how hydrogen atoms in ammonia are replaced with one, two, or three alkyl groups to form primary, secondary, and tertiary amines respectively.

[Generated by AI]

Figure 10 shows examples of a primary, secondary and tertiary amine. Can you deduce which is which? Check your answer below when you are ready (note that the naming of amines is not required: the names are included here for completeness).

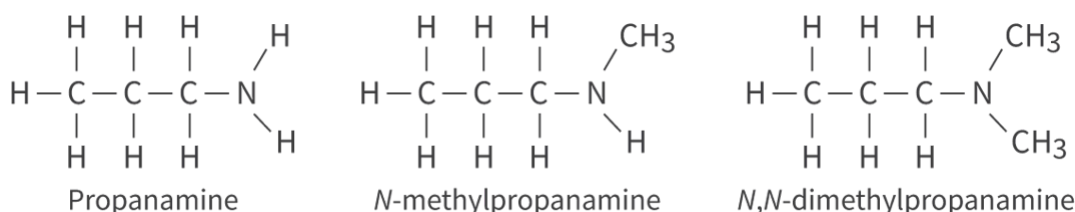


Figure 10. Can you deduce which of these compounds is a primary, secondary and tertiary amine?

 More information for figure 10

From left to right, the image shows the chemical structures of three amines: \n\n1. **Propanamine:** A primary amine consisting of a three-carbon chain. The nitrogen atom (N) is bonded to the last carbon in the chain with two hydrogen atoms attached to it. The structure is linear with bonds shown between the carbons and between carbon and nitrogen.\n\n2. **N-methylpropanamine:** A secondary amine, similar to propanamine but with an additional methyl group (CH₃) attached to the nitrogen atom. This modifies the nitrogen, giving it only one hydrogen atom directly bonded. The structure remains linear.\n\n3. **N,N-dimethylpropanamine:** A tertiary amine with the same three carbon chain. The nitrogen atom has two methyl groups (CH₃) attached to it, replacing the hydrogen atoms completely. The nitrogens in these structures are shown with clear bonds to the rest of the components representing each of their attachments.\n\nThese structures illustrate the progression from primary to tertiary amines, based on the number of carbon-containing groups attached to the nitrogen atom.

[Generated by AI]

Propanamine is a primary amine because it has only one alkyl group bonded to the nitrogen of the amino group. *N*-methylpropanamine is a secondary amine because it has two alkyl groups bonded to the nitrogen of the amino group. And *N,N*-dimethylpropanamine is a tertiary amine because it has three alkyl groups bonded to the nitrogen of the amino group.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. Sketch the full structural formulas of the two position isomers with the molecular formula C_4H_9OH . Name the isomers using IUPAC nomenclature rules.
2. Sketch the full structural formulas of the two functional groups isomers with the molecular formula C_2H_6O . Name the isomers using IUPAC nomenclature rules.
3. Sketch the full structural formulas of the three chain isomers with the molecular formula C_5H_{12} . Name the isomers using IUPAC nomenclature rules.
4. State the type of structural isomerism shown by the two isomers in **Figure 11** and the functional group of each.

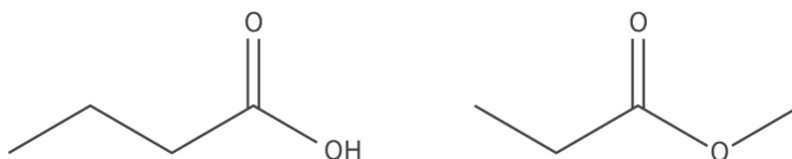


Figure 11. Which structural isomers are shown?

 More information for figure 11

The image shows two structural isomers in a diagram format. On the left, there is a molecule with a carboxylic acid group as the functional group, which is depicted by the COOH structure at the terminal end of the carbon chain. On the right, there is a molecule with an ester group functional group, characterized by the COO structure in the main chain. These represent two different types of structural isomerism in organic chemistry, one being a carboxylic acid and the other an ester, showing how the arrangement of atoms results in different functional groups and chemical properties.

[Generated by AI]

5. State the type of structural isomerism shown by the three isomers $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHBrCH}_3$ and $\text{CH}_3\text{CH}_2\text{CHBrCH}_2\text{CH}_3$. Name each isomer using IUPAC nomenclature rules.
6. State the type of structural isomerism shown by the two isomers in **Figure 12**. Name each isomer using IUPAC nomenclature rules.

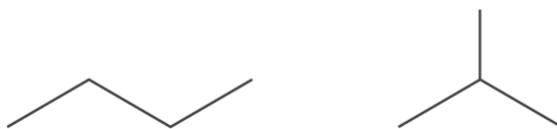


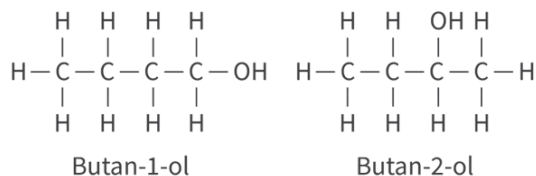
Figure 12. Which structural isomers are shown?

 More information for figure 12

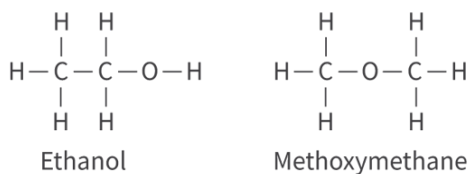
The image shows two chemical structures representing structural isomers. On the left, there is a chain of five carbon atoms arranged in a zigzag pattern, indicating a linear hydrocarbon structure. This structure does not have any branches, resembling a straight line. On the right, there is a central carbon atom with three other carbon atoms branching out in a Y-shaped arrangement, suggesting a branched hydrocarbon. These two structures represent two structural isomers differing in their carbon chain arrangement.

[Generated by AI]

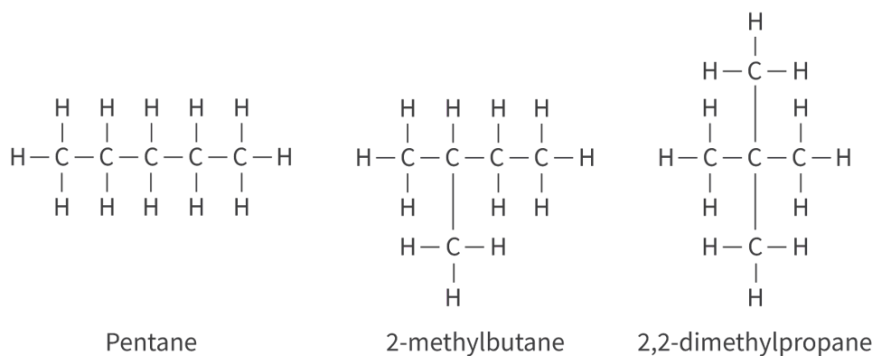
1. Sketch the full structural formulas of the two position isomers with the molecular formula $\text{C}_4\text{H}_9\text{OH}$. Name the isomers using IUPAC nomenclature rules.



2. Sketch the full structural formulas of the two functional groups isomers with the molecular formula $\text{C}_2\text{H}_6\text{O}$. Name the isomers using IUPAC nomenclature rules.

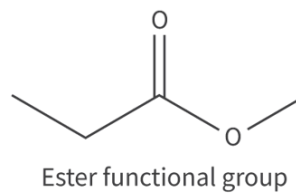
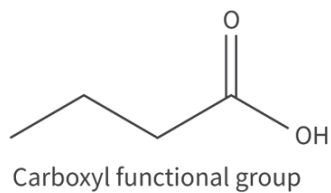


3. Sketch the full structural formulas of the three chain isomers with the molecular formula C_5H_{12} . Name the isomers using IUPAC nomenclature rules.



4. State the type of structural isomerism shown by the two isomers below.

The type of isomerism is functional group isomerism.



5. State the type of structural isomerism shown by the three isomers $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHBrCH}_3$ and $\text{CH}_3\text{CH}_2\text{CHBrCH}_2\text{CH}_3$. Name each isomer using IUPAC nomenclature rules.

The type of isomerism is position isomerism.

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ is 1-bromopentane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHBrCH}_3$ is 2-bromopentane

$\text{CH}_3\text{CH}_2\text{CHBrCH}_2\text{CH}_3$ is 3-bromopentane

6. State the type of structural isomerism shown by the two isomers below. Name each isomer using IUPAC nomenclature rules.

The type of isomerism is chain isomerism.

